

Replacement Classes and the Isometry Groups of Combinatorial Banach Spaces

A research note on the questions of Brech–Ferenczi–Tcaciuc

21 June 2026

Abstract

For a family $\mathcal{F} \subseteq [\mathbb{N}]^{<\omega}$, define the replacement relation $i \preceq_{\mathcal{F}} j$ by requiring that j may replace i in every finite context without destroying membership in $\mathcal{F}$. We prove that this relation is a preorder for every family and is total whenever $\mathcal{F}$ is spreading. Its equivalence classes are consecutive intervals of $\mathbb{N}$, and the full permutation automorphism group of a spreading family is exactly the unrestricted product of the symmetric groups on those intervals. Combining this with the signed-permutation theorem of Brech–Ferenczi–Tcaciuc gives a complete intrinsic description of the surjective linear isometry group of every regular combinatorial Banach space and of its dual. We also classify which permutation groups occur, verify the supplied threshold-family result, and obtain a short rigidity theorem: two permutation-isomorphic spreading families on $\mathbb{N}$ are equal. A literature search found close finite antecedents through desirability and vicinal preorders, but no exact statement of the countably infinite Banach-space conclusion; this bibliographic observation is not a formal novelty certification.

1 Problem, verdict on the partial result, and main conclusion

Throughout, $\mathbb{N} = \{1, 2, \dots\}$. A family means a subset of $[\mathbb{N}]^{<\omega}$. It is *spreading* if every coordinatewise increase of a member remains in the family. It is *regular* if it is hereditary, spreading, compact as a subspace of $2^{\mathbb{N}}$, and contains all singletons. For a regular family $\mathcal{F}$, the associated combinatorial Banach space is the completion of c_{00} under

$$\|x\|_{\mathcal{F}} = \sup_{A \in \mathcal{F}} \sum_{n \in A} |x(n)|.$$

Brech, Ferenczi, and Tcaciuc proved that every surjective linear isometry of a combinatorial space, or of its dual, is a signed permutation of the canonical basis. They then asked which permutations preserve a given regular family and for which combinatorial spaces the full isometry group can be explicitly described [2]. The same problem was still listed in the 2022 survey of Antunes and Beanland [1, Problem 3.1].

The supplied partial result treats

$$\mathcal{F}_g = \{\emptyset\} \cup \{A \in [\mathbb{N}]^{<\omega} : A \neq \emptyset, |A| \leq g(\min A)\},$$

where $g : \mathbb{N} \rightarrow \mathbb{N}$ is nondecreasing and $g \geq 1$. Its conclusion is that $\text{Aut}(\mathcal{F}_g)$ consists precisely of the permutations preserving every level set of g .

Remark 1.1 (Verdict on the supplied argument). The partial result is correct. In particular, its compactness proof, the invariant

$$h_{\mathcal{F}_g}(n) = \max\{|A| : A \in \mathcal{F}_g, n \in A\} = g(n),$$

and the converse argument for level-preserving permutations are all sound. Section 5 gives a self-contained verification. The main point of the present note is that the same replacement idea, formulated at the right level of generality, settles the permutation problem for every spreading family.

Here is the resulting answer. The canonical partition appearing in the statement is defined in Section 2.

Theorem 1.2 (Main isometry theorem). *Let $\mathcal{F}$ be a regular family, let $\mathcal{C}_{\mathcal{F}}$ be its replacement classes, and let $\mathbb{U}_{\mathbb{K}} = \{-1, 1\}$ in the real case and $\mathbb{U}_{\mathbb{K}} = \{z \in \mathbb{C} : |z| = 1\}$ in the complex case. A map $T : X_{\mathcal{F}} \rightarrow X_{\mathcal{F}}$ is a surjective linear isometry if and only if there are $\varepsilon = (\varepsilon_n)_{n \in \mathbb{N}} \in \mathbb{U}_{\mathbb{K}}^{\mathbb{N}}$ and a permutation π satisfying $\pi(C) = C$ for every $C \in \mathcal{C}_{\mathcal{F}}$ such that*

$$Te_n = \varepsilon_n e_{\pi(n)} \quad (n \in \mathbb{N}).$$

The analogous statement holds on $X_{\mathcal{F}}^$. Consequently,*

$$\text{Isom}(X_{\mathcal{F}}) \cong \mathbb{U}_{\mathbb{K}}^{\mathbb{N}} \rtimes \prod_{C \in \mathcal{C}_{\mathcal{F}}} \text{Sym}(C),$$

with the same permutation part for $\text{Isom}(X_{\mathcal{F}}^)$.*

The product in this formula is the unrestricted direct product: independent permutations may be chosen on all replacement classes, including an infinite class. The classes are consecutive intervals, so the permutation part is a Young subgroup associated with an interval partition of $\mathbb{N}$.

2 The replacement preorder

The relation below is the set-system analogue of Isbell's desirability relation for simple games [6]; closely related generalized vicinal preorders occur in the theory of shifted simplicial complexes [8]. We include all details because two elementary features of the relation do the entire job.

Definition 2.1 (Replacement preorder). For a family $\mathcal{F} \subseteq [\mathbb{N}]^{<\omega}$ and $i, j \in \mathbb{N}$, write $i \preceq_{\mathcal{F}} j$ if

$$A \cup \{i\} \in \mathcal{F} \implies A \cup \{j\} \in \mathcal{F} \quad (2.1)$$

for every finite $A \subseteq \mathbb{N} \setminus \{i, j\}$. Thus j can replace i in every context. Write

$$i \sim_{\mathcal{F}} j \iff i \preceq_{\mathcal{F}} j \text{ and } j \preceq_{\mathcal{F}} i.$$

The equivalence classes of $\sim_{\mathcal{F}}$ will be called the *replacement classes* of $\mathcal{F}$ and denoted by $\mathcal{C}_{\mathcal{F}}$.

Proposition 2.2. *For every family $\mathcal{F}$, the relation $\preceq_{\mathcal{F}}$ is a preorder. If π is a permutation and $\mathcal{G} = \pi[\mathcal{F}]$, then*

$$i \preceq_{\mathcal{F}} j \iff \pi(i) \preceq_{\mathcal{G}} \pi(j). \quad (2.2)$$

Proof. Reflexivity is immediate. For transitivity, the cases with repeated indices are immediate, so suppose that i, j, k are pairwise distinct, $i \preceq_{\mathcal{F}} j$, and $j \preceq_{\mathcal{F}} k$. Let $A \subseteq \mathbb{N} \setminus \{i, k\}$ be finite and suppose $A \cup \{i\} \in \mathcal{F}$.

If $j \notin A$, replace i by j and then j by k . This gives $A \cup \{k\} \in \mathcal{F}$. If $j \in A$, put $B = A \setminus \{j\}$. Starting from $B \cup \{i, j\} \in \mathcal{F}$, first use $j \preceq_{\mathcal{F}} k$ with context $B \cup \{i\}$ to obtain $B \cup \{i, k\} \in \mathcal{F}$. Then use $i \preceq_{\mathcal{F}} j$ with context $B \cup \{k\}$ to obtain $B \cup \{j, k\} = A \cup \{k\} \in \mathcal{F}$. Thus $i \preceq_{\mathcal{F}} k$.

For (2.2), apply π^{-1} to a context avoiding $\pi(i), \pi(j)$, use (2.1) in $\mathcal{F}$, and apply π again. The reverse implication follows in the same way from π^{-1} . $\square$

Proposition 2.3. *If $\mathcal{F}$ is spreading, then:*

- (i) *$i < j$ implies $i \preceq_{\mathcal{F}} j$, so the replacement preorder is total;*
- (ii) *each replacement class is a consecutive interval of $\mathbb{N}$;*
- (iii) *for $i \neq j$, one has $i \sim_{\mathcal{F}} j$ if and only if the transposition $(i\ j)$ belongs to $\text{Aut}(\mathcal{F})$.*

Proof. For (i), if $A \cup \{i\} \in \mathcal{F}$ and A avoids i, j , the map that fixes A and sends i to j is an injective pointwise increase. The spreading property therefore gives $A \cup \{j\} \in \mathcal{F}$.

For (ii), suppose $i < k < j$ and $i \sim_{\mathcal{F}} j$. By (i), $i \preceq_{\mathcal{F}} k \preceq_{\mathcal{F}} j$. Combining these inequalities with $j \preceq_{\mathcal{F}} i$ and using transitivity gives both $i \sim_{\mathcal{F}} k$ and $k \sim_{\mathcal{F}} j$.

For (iii), if the transposition is an automorphism, applying it to $A \cup \{i\}$ or $A \cup \{j\}$ proves the two replacement implications. Conversely, if $i \sim_{\mathcal{F}} j$, the transposition fixes every finite set that contains both or neither of i, j ; when exactly one is present, the relevant replacement implication preserves membership. Hence the transposition preserves $\mathcal{F}$ in both directions. $\square$

The quotient $\mathbb{N}/\sim_{\mathcal{F}}$ is ordered by

$$[i]_{\mathcal{F}} \leq [j]_{\mathcal{F}} \iff i \preceq_{\mathcal{F}} j. \quad (2.3)$$

This is well defined by Proposition 2.2 and is a linear order when $\mathcal{F}$ is spreading. More is true: it is a well-order. Indeed, for any nonempty collection of classes, the class containing the least integer in their union is its least member. Equivalently, the replacement classes form either a finite interval partition of $\mathbb{N}$ or a sequence of finite consecutive intervals indexed by $\mathbb{N}$.

3 Automorphisms of every spreading family

Theorem 3.1 (Automorphism theorem). *Let $\mathcal{F} \subseteq [\mathbb{N}]^{<\omega}$ be spreading. Then*

$$\text{Aut}(\mathcal{F}) = \{\pi \in \text{Sym}(\mathbb{N}) : \pi(C) = C \text{ for every } C \in \mathcal{C}_{\mathcal{F}}\} \cong \prod_{C \in \mathcal{C}_{\mathcal{F}}} \text{Sym}(C). \quad (3.1)$$

In particular, the replacement classes are exactly the orbits of $\text{Aut}(\mathcal{F})$ on $\mathbb{N}$.

Proof. Let $\pi \in \text{Aut}(\mathcal{F})$. Proposition 2.2 shows that π preserves the replacement preorder and hence induces an order automorphism of the well-ordered quotient $\mathbb{N}/\sim_{\mathcal{F}}$. A well-order has no nonidentity order automorphism, so every replacement class is fixed setwise. This proves the inclusion from left to right in (3.1).

Conversely, suppose that $\pi(C) = C$ for every replacement class. Fix a finite set $A \subseteq \mathbb{N}$. For each of the finitely many classes C meeting A , the finite partial bijection

$$\pi|_{A \cap C} : A \cap C \longrightarrow \pi(A \cap C)$$

extends to a permutation σ_C of C with finite support. One explicit construction is to view the partial bijection on the finite set $(A \cap C) \cup \pi(A \cap C)$ as a disjoint union of directed paths and cycles and close each path into a cycle. Let σ agree with these σ_C and be the identity elsewhere. Then σ has finite support and is a finite product of transpositions inside replacement classes. By Proposition 2.3(iii), every such transposition is an automorphism of $\mathcal{F}$, so $\sigma \in \text{Aut}(\mathcal{F})$. Since $\sigma(A) = \pi(A)$,

$$A \in \mathcal{F} \iff \pi(A) \in \mathcal{F}.$$

This holds for every finite A , hence $\pi \in \text{Aut}(\mathcal{F})$.

Finally, arbitrary permutations inside a class are allowed, so each class is one orbit, while the first part shows that no orbit meets two classes. $\square$

Corollary 3.2 (Block-count normal form). *Let $\mathcal{F}$ be spreading. If finite sets $A, B \subseteq \mathbb{N}$ satisfy*

$$|A \cap C| = |B \cap C| \quad (C \in \mathcal{C}_{\mathcal{F}}),$$

then $A \in \mathcal{F}$ if and only if $B \in \mathcal{F}$.

Proof. Choose, independently in each replacement class, a permutation sending $A \cap C$ to $B \cap C$, and combine them. The resulting block-preserving permutation is an automorphism by Theorem 3.1. $\square$

Thus every spreading family is completely symmetric inside its replacement blocks. All asymmetry is encoded by the ordered list of blocks and by which finite block-count vectors are admitted.

4 Consequences for combinatorial Banach spaces

We now specialize to regular families and use the signed-permutation theorem of Brech–Ferenczi–Tcaciuc. Their result says that every surjective linear isometry between dual combinatorial spaces has the form $e_n^* \mapsto \varepsilon_n e_{\pi(n)}^*$, and their corollary says that the permutation π is admissible exactly when it carries one defining family onto the other. The corresponding assertion for the primal spaces follows by adjoints [2, Theorem 10 and Corollary 12].

Proof of Theorem 1.2. By the cited theorem, a surjective linear isometry of $X_{\mathcal{F}}$ or $X_{\mathcal{F}}^*$ is a signed permutation, and its permutation part belongs to $\text{Aut}(\mathcal{F})$. Theorem 3.1 identifies this automorphism group with $\prod_{C \in \mathcal{C}_{\mathcal{F}}} \text{Sym}(C)$. Conversely, every coordinatewise sign change is an isometry because the canonical basis is 1-unconditional, and every block-preserving permutation preserves $\mathcal{F}$, hence the defining norm. The displayed semidirect product uses the natural action of coordinate permutations on sign sequences. $\square$

This gives a universal intrinsic answer to the two questions in [2]: for every regular family, compute the replacement equivalence classes, and then permit arbitrary permutations within each class and no permutations between classes.

5 Verification and extension of the threshold-family result

Proposition 5.1 (Threshold families). *Let $g : \mathbb{N} \rightarrow \mathbb{N}$ be nondecreasing with $g(n) \geq 1$, and define*

$$\mathcal{F}_g = \{\emptyset\} \cup \{A \in [\mathbb{N}]^{<\omega} : A \neq \emptyset, |A| \leq g(\min A)\}.$$

Then $\mathcal{F}_g$ is regular and

$$\text{Aut}(\mathcal{F}_g) = \{\pi \in \text{Sym}(\mathbb{N}) : g(\pi(n)) = g(n) \text{ for all } n\}. \quad (5.1)$$

Its replacement classes are exactly the nonempty level sets $L_m = \{n : g(n) = m\}$.

Proof. If $B \subseteq A \in \mathcal{F}_g$ is nonempty, then $|B| \leq |A| \leq g(\min A) \leq g(\min B)$, so $\mathcal{F}_g$ is hereditary. If $A = \{n_1 < \dots < n_k\} \in \mathcal{F}_g$ and $m_r \geq n_r$ with $m_1 < \dots < m_k$, then $k \leq g(n_1) \leq g(m_1)$, so $\mathcal{F}_g$ is spreading. All singletons belong to it.

For compactness, it is enough to prove that $\mathcal{F}_g$ is closed in the compact metrizable space $2^{\mathbb{N}}$. Suppose $A_r \in \mathcal{F}_g$ converges pointwise to $A \subseteq \mathbb{N}$. If A were infinite and $a = \min A$, then eventually A_r would contain the first $g(a) + 1$ members of A (including a), while containing no integer below a . Hence $\min A_r = a$ and $|A_r| \geq g(a) + 1$, a contradiction. Thus A is finite. If $A \neq \emptyset$ and

$a = \min A$, then eventually every member of A lies in A_r and no integer below a does. Consequently $|A| \leq |A_r| \leq g(a)$, so $A \in \mathcal{F}_g$. The empty set is already included.

For $n \in \mathbb{N}$, set

$$h_{\mathcal{F}_g}(n) = \max\{|A| : A \in \mathcal{F}_g, n \in A\}.$$

Every member containing n has size at most $g(\min A) \leq g(n)$, while $\{n, n+1, \dots, n+g(n)-1\} \in \mathcal{F}_g$. Therefore $h_{\mathcal{F}_g}(n) = g(n)$. Since a family automorphism preserves the cardinalities of members containing a given point, every automorphism preserves g .

Conversely, suppose $g \circ \pi = g$. If $A \in \mathcal{F}_g$ is nonempty, let $r = \min A$, and choose $s \in A$ with $\pi(s) = \min \pi(A)$. Since $s \geq r$,

$$|\pi(A)| = |A| \leq g(r) \leq g(s) = g(\pi(s)) = g(\min \pi(A)).$$

Thus $\pi(A) \in \mathcal{F}_g$, and the same argument for π^{-1} gives equality of the families. This proves (5.1).

The level sets are the orbits of this automorphism group. By Theorem 3.1, the orbits are exactly the replacement classes. $\square$

The proposition strictly contains the step-function example in [2, Example 18]: the threshold may have arbitrary finite or infinite plateaus and may jump by more than one. Constant g gives the full symmetric permutation group, while $g(n) = n$ gives the first Schreier family and only singleton replacement classes.

6 Exactly which permutation groups occur?

Call $(I_r)_{r \in R}$ an *interval partition* of $\mathbb{N}$ if R is either $\{1, \dots, q\}$ or $\mathbb{N}$, the sets I_r are nonempty consecutive intervals, $I_r < I_s$ for $r < s$, and $\bigcup_r I_r = \mathbb{N}$.

Theorem 6.1 (Realization and classification). *As concrete subgroups of $\text{Sym}(\mathbb{N})$, the permutation parts of the isometry groups of regular combinatorial spaces are exactly the interval Young subgroups*

$$\prod_{r \in R} \text{Sym}(I_r) \tag{6.1}$$

associated with interval partitions of $\mathbb{N}$.

Proof. For every regular family, Proposition 2.3 and Theorem 3.1 show that its automorphism group has the form (6.1).

Conversely, fix an interval partition $(I_r)_{r \in R}$ and define $g(n) = r$ for $n \in I_r$. Then g is nondecreasing and at least one. Proposition 5.1 shows that $\mathcal{F}_g$ is regular, its replacement classes are precisely the I_r , and its automorphism group is exactly (6.1). $\square$

If the partition has finitely many blocks, its last block is necessarily an infinite tail. If it has infinitely many blocks, every block is finite. The full isometry group is obtained from (6.1) by adjoining all coordinatewise signs as in Theorem 1.2.

7 Permutation rigidity of spreading families

Brech and Piña proved that hereditary spreading families on $\mathbb{N}$ which are images of one another under a permutation must be equal [3, Theorem 14]. The replacement-class theorem gives a short proof and shows that heredity is unnecessary.

Theorem 7.1 (Rigidity). *Let $\mathcal{F}, \mathcal{G} \subseteq [\mathbb{N}]^{<\omega}$ be spreading. If $\mathcal{G} = \pi[\mathcal{F}]$ for a permutation π of $\mathbb{N}$, then $\mathcal{F} = \mathcal{G}$.*

Proof. By Proposition 2.2, π transports the replacement preorder of $\mathcal{F}$ to that of $\mathcal{G}$. Hence it maps the first replacement class of $\mathcal{F}$ onto the first replacement class of $\mathcal{G}$, the second onto the second, and so on, preserving the cardinality of each class.

An ordered interval partition of $\mathbb{N}$ is determined by this ordered list of cardinalities. Indeed, the first blocks are initial intervals of the same size and hence coincide. Inductively, after the preceding blocks have been identified, the next blocks begin at the same least unused integer and have the same size, so they coincide as well. If one block is infinite, it is the final tail and the argument terminates.

Thus $\mathcal{F}$ and $\mathcal{G}$ have the same replacement classes, and π preserves each of them setwise. Theorem 3.1 then gives $\pi \in \text{Aut}(\mathcal{F})$. Therefore $\mathcal{G} = \pi[\mathcal{F}] = \mathcal{F}$. $\square$

Corollary 7.2. *If $\mathcal{F}$ and $\mathcal{G}$ are regular and $X_{\mathcal{F}}$ is linearly isometric to $X_{\mathcal{G}}$ (equivalently, their duals are linearly isometric), then $\mathcal{F} = \mathcal{G}$. After this identification, every isometry has the blockwise signed form of Theorem 1.2.*

Proof. Brech–Ferenczi–Tcaciuc reduce an isometry to a permutation satisfying $\mathcal{G} = \pi[\mathcal{F}]$ [2, Corollary 12]; apply Theorem 7.1. This also recovers the regular-family Banach–Stone conclusion of [3, Corollary 15]. $\square$

8 Literature status and scope

The replacement relation itself is not new in spirit. Isbell introduced the desirability relation for finite simple games [6]; complete simple games are those for which the resulting preorder is total, and their player types are its equivalence classes [4]. Carreras and Owen studied automorphisms in that setting [5]. Klivans used a generalized vicinal preorder to characterize finite shifted simplicial complexes by totality [8]. Kalai also proved a finite rigidity statement for shifted families in the stronger language of weak isomorphism [7, Proposition 4.2].

On the Banach-space side, Brech–Ferenczi–Tcaciuc supplied the decisive signed-permutation reduction [2]; Brech–Piña later established the rigidity of hereditary spreading families under permutation [3]; and the 2022 survey still recorded the explicit-isometry-group problem [1].

A targeted search of these references, their citation trails, arXiv, and recent work through 21 June 2026 did not locate the exact statement of Theorems 3.1 and 1.2 for countably infinite spreading families and combinatorial Banach spaces. The finite game-theoretic and shifted-complex literature is a substantial antecedent, so this search result should be read only as a careful status report, not as a guarantee of novelty.

9 Conclusion

The supplied threshold argument was correct, but its invariant was revealing a more general object: the replacement preorder. Spreading makes that preorder total, the well-order of $\mathbb{N}$ prevents automorphisms from moving its classes, and mutual replaceability makes every permutation inside a class legal. This yields the complete formula

$$\text{Aut}(\mathcal{F}) = \prod_{C \in \mathcal{C}_{\mathcal{F}}} \text{Sym}(C)$$

for every spreading family and, after the theorem of Brech–Ferenczi–Tcaciuc, the full signed formula for every regular combinatorial Banach space and its dual.

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